

Exercises

Design principles & Patterns

Exercise 1: Implementing the Singleton Pattern

```
Singleton.java
import java.util.*;
class Singleton {
    private static Singleton instance;
    Singleton() {
        System.out.println("Created a Singleton Class");
    } //Private constructor
    public static Singleton getInstance(){
        if (instance==null){
            instance=new Singleton();
        }
        return instance;
    }
}
```

PORTS

▼ TERMINAL

```
● PS P:\Cognizant_exercise> javac Singleton.java Main.java
● PS P:\Cognizant_exercise> java Main
Created a Singleton Class
Both are same
❖ PS P:\Cognizant_exercise> █
```

Main Loading...

```
1 public class Main{
2     public static void main(String args[]){
3         Singleton s1=Singleton.getInstance();
4         Singleton s2=Singleton.getInstance();
5         if(s1==s2){
6             System.out.println("Both are same");
7         }
8     }
9 }
```

Exercise 2: Implementing the Factory Method Pattern

```
FactoryMethod.java
1  import java.util.*;
2  interface Animal{
3      void speak();
4  }
5  class Dog implements Animal{
6      public void speak(){
7          System.out.println("Woof");
8      }
9  }
10 class Cat implements Animal{
11     public void speak(){
12         System.out.println("Meow");
13     }
14 }
15 class FactoryMethod {
16     public static void getFactory(String type){
17         if(type=="Dog"){
18             Dog d=new Dog();
19         }
20         else if(type=="Cat"){
21             Cat c=new Cat();
22         }
23         else{
24             System.out.println("Invalid Type");
25         }
26     }
27 }
```

```
public class Main{  
    public static void main(String args[]){x:  
        Animal a=new Dog();  
        a.speak();  
        Animal b=new Cat();  
        b.speak();  
    }  
}
```

AL PORTS QUERY RESULTS

✓ **TERMINAL**

```
● PS P:\Cognizant_exercise> javac FactoryMethod.java Main.java  
● PS P:\Cognizant_exercise> java Main  
Woof  
Meow  
❖ PS P:\Cognizant_exercise> 
```

Exercise 2: E-commerce Platform Search Function

```
import java.util.*;

class Ecommerce {
    static class Product{
        int id;
        String pname;
        String Category;
        double price;
        Product(int id,String pname,String Category,double price){
            this.id=id;
            this.pname=pname;
            this.Category=Category;
            this.price=price;
        }
        public String toString(){
            return id + "-" + pname + "-" + Category + "Rs."+price;
        }
    }
    //It follows Interface Segregation Principle.
    interface SearchStrategy{
        List<Product> search(List<Product> products, String query);
    }
    //It follows Single Responsibility Principle as it has only one method to implement in the class which implements this
    static class KeywordSearch implements SearchStrategy{
```

```
        public List<Product> search(List<Product> products, String query){
            List<Product> result=new ArrayList<>();
            for(Product p:products){
                if(p.pname.toLowerCase().contains(query.toLowerCase())){
                    result.add(p);
                }
            }
            return result;
        }
    }
    static class PriceSearch implements SearchStrategy{
        public List<Product> search(List<Product> products, String query){
            List<Product> result=new ArrayList<>();
            double price=Double.parseDouble(query);
            for(Product p:products){
                if(p.price<=price){
                    result.add(p);
                }
            }
            return result;
        }
    }
}
```

```

}
static class ProductCatalog{
    private List<Product> products=new ArrayList<>();
    private Map<Integer,Product> productMap=new HashMap<>();
    void addProduct(Product p){
        products.add(p);
        productMap.put(p.id,p);
    }
    Product getById(int id){
        return productMap.get(id);
    }
    List<Product> search(SearchStrategy strategy,String query){
        return strategy.search(products,query);
    }
}

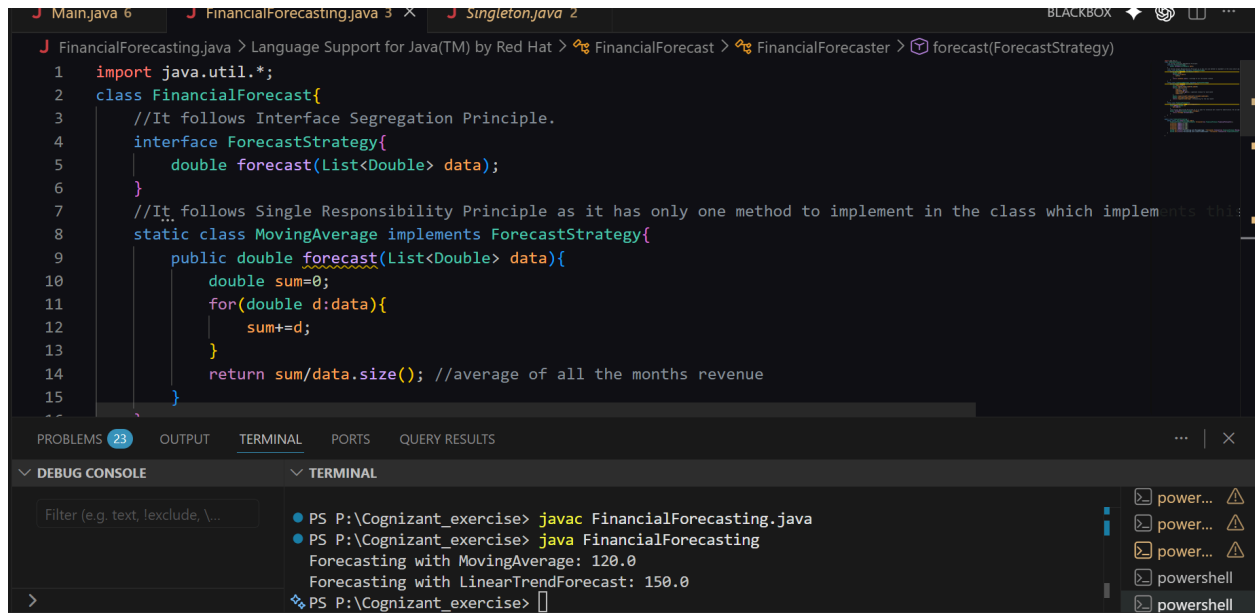
```

```

public class Main{
    Run main | Debug main | Run | Debug
    public static void main(String[] args){
        Ecommerce.ProductCatalog catalog=new Ecommerce.ProductCatalog();
        catalog.addProduct(new Ecommerce.Product(id: 1,pname: "Laptop",Category: "Electronics",price: 50000));
        catalog.addProduct(new Ecommerce.Product(id: 2,pname: "Mobile",Category: "Electronics",price: 20000));
        catalog.addProduct(new Ecommerce.Product(id: 3,pname: "Shirt",Category: "Clothing",price: 1000));
        catalog.addProduct(new Ecommerce.Product(id: 4,pname: "Jeans",Category: "Clothing",price: 2000));
        catalog.addProduct(new Ecommerce.Product(id: 5,pname: "Shoes",Category: "Footwear",price: 3000));
        System.out.println(x: "Search by keyword 'Laptop':");
        List<Ecommerce.Product> result=catalog.search(new Ecommerce.KeywordSearch(),query: "Laptop");
        for(Ecommerce.Product p:result){
            System.out.println(p);
        }
        System.out.println(x: "Search by price <= 2000:");
        result=catalog.search(new Ecommerce.PriceSearch(),query: "2000");
        for(Ecommerce.Product p:result){
            System.out.println(p);
        }
    }
}

```

Exercise 7: Financial Forecasting



```
1 import java.util.*;
2 class FinancialForecast{
3     //It follows Interface Segregation Principle.
4     interface ForecastStrategy{
5         double forecast(List<Double> data);
6     }
7     //It follows Single Responsibility Principle as it has only one method to implement in the class which implements this
8     static class MovingAverage implements ForecastStrategy{
9         public double forecast(List<Double> data){
10             double sum=0;
11             for(double d:data){
12                 sum+=d;
13             }
14             return sum/data.size(); //average of all the months revenue
15         }
16     }
17 }
```

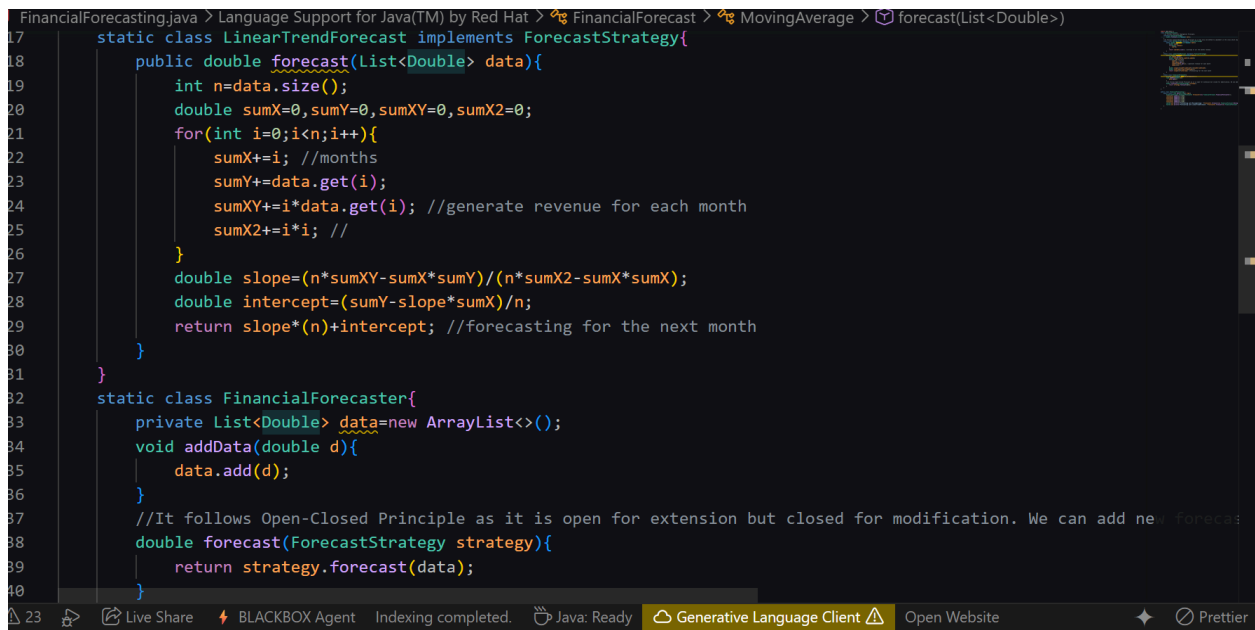
PROBLEMS 23 OUTPUT TERMINAL PORTS QUERY RESULTS

DEBBUG CONSOLE TERMINAL

Filter (e.g. text, exclude, ...)

- PS P:\Cognizant_exercise> javac FinancialForecasting.java
- PS P:\Cognizant_exercise> java FinancialForecasting
- Forecasting with MovingAverage: 120.0
- Forecasting with LinearTrendForecast: 150.0
- PS P:\Cognizant_exercise>

power... power... power... powershell powershell



```
17 static class LinearTrendForecast implements ForecastStrategy{
18     public double forecast(List<Double> data){
19         int n=data.size();
20         double sumX=0,sumY=0,sumXY=0,sumX2=0;
21         for(int i=0;i<n;i++){
22             sumX+=i; //months
23             sumY+=data.get(i);
24             sumXY+=i*data.get(i); //generate revenue for each month
25             sumX2+=i*i; //
26         }
27         double slope=(n*sumXY-sumX*sumY)/(n*sumX2-sumX*sumX);
28         double intercept=(sumY-slope*sumX)/n;
29         return slope*(n)+intercept; //forecasting for the next month
30     }
31 }
32 static class FinancialForecaster{
33     private List<Double> data=new ArrayList<>();
34     void addData(double d){
35         data.add(d);
36     }
37     //It follows Open-Closed Principle as it is open for extension but closed for modification. We can add new forecast
38     double forecast(ForecastStrategy strategy){
39         return strategy.forecast(data);
40     }
41 }
```

FinancialForecasting.java > Language Support for Java(TM) by Red Hat > FinancialForecast > MovingAverage > forecast(List<Double>)

23 Live Share BLACKBOX Agent Indexing completed. Java: Ready Generative Language Client Open Website Prettier

```
    }  
    }  
}  
  
public class FinancialForecasting {  
    Run main | Debug main | Run | Debug  
    public static void main(String[] args){  
        FinancialForecast.FinancialForecaster forecaster=new FinancialForecast.FinancialForecaster();  
        forecaster.addData(d: 100);  
        forecaster.addData(d: 110);  
        forecaster.addData(d: 120);  
        forecaster.addData(d: 130);  
        forecaster.addData(d: 140);  
        System.out.println("Forecasting with MovingAverage: "+forecaster.forecast(new FinancialForecast.MovingAverage()));  
        System.out.println("Forecasting with LinearTrendForecast: "+forecaster.forecast(new FinancialForecast.LinearTrendForecast()));  
    }  
}
```

Exercise 1: Control Structures

Scenario 1: The bank wants to apply a discount to loan interest rates for customers above 60 years old.

Statement details

```
DECLARE
  CURSOR c_customers IS
    SELECT customer_id, age
    FROM customers;
BEGIN
  FOR cust_rec IN c_customers LOOP
    IF cust_rec.age > 60 THEN
      UPDATE loans
      SET interest_rate = interest_rate - (interest_rate * 0.01)
      WHERE customer_id = cust_rec.customer_id;

      DBMS_OUTPUT.PUT_LINE('Discount applied for Customer ID: ' ||
cust_rec.customer_id);
    END IF;
  END LOOP;

  COMMIT;
END;
```

 Insert Into Editor

Result

```
Discount applied for Customer ID: 101
Discount applied for Customer ID: 103

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.006
```


Scenario 2: A customer can be promoted to VIP status based on their balance.

Executed on: Jun 30, 2026, 7:17:21 PM

```
DECLARE
  CURSOR c_customers IS
    SELECT customer_id, balance
    FROM customers;
BEGIN
  FOR cust_rec IN c_customers LOOP
    IF cust_rec.balance > 10000 THEN
      UPDATE customers
      SET is_vip = 'Y'
      WHERE customer_id = cust_rec.customer_id;
      DBMS_OUTPUT.PUT_LINE('Customer ID ' || cust_rec.customer_id ||
        ' marked as VIP');
    END IF;
  END LOOP;
  COMMIT;
END;
```

 Insert Into Editor

Result

```
Customer ID 101 marked as VIP
Customer ID 102 marked as VIP
```

Scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.

```
DECLARE
  CURSOR c_due_loans IS
    SELECT loan_id, customer_id, due_date
    FROM loans
    WHERE due_date BETWEEN SYSDATE AND (SYSDATE + 30);
BEGIN
  FOR loan_rec IN c_due_loans LOOP
    DBMS_OUTPUT.PUT_LINE(
      'Reminder: Loan ID ' || loan_rec.loan_id ||
      ' for Customer ID ' || loan_rec.customer_id ||
      ' is due on ' || TO_CHAR(loan_rec.due_date, 'DD-MON-YYYY')
    );
  END LOOP;
END;
```

 Insert Into Editor

Result

```
Reminder: Loan ID 1 for Customer ID 101 is due on 15-JUL-2026
Reminder: Loan ID 3 for Customer ID 103 is due on 10-JUL-2026
```

Exercise 3: Stored Procedures

Scenario 1: The bank needs to process monthly interest for all savings accounts.

Executed on: Jun 30, 2026, 7:54:58 PM

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
AS
BEGIN
    UPDATE savings_accounts
    SET balance = balance+(balance*0.01);
    DBMS_OUTPUT.PUT_LINE('Monthly interest applied to ' ||
SQL%ROWCOUNT || ' accounts.');
```

 Insert Into Editor

Result

Procedure PROCESSMONTHLYINTEREST compiled




Scenario 2: The bank wants to implement a bonus scheme for employees based on their performance.

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus (  
    p_department_id IN NUMBER,  
    p_bonus_pct     IN NUMBER  
)  
AS  
BEGIN  
    UPDATE employees  
    SET salary=salary+(salary*p_bonus_pct/100)  
    WHERE department_id=p_department_id;  
    IF SQL%ROWCOUNT=0 THEN  
        DBMS_OUTPUT.PUT_LINE('No employees found in department ' ||  
p_department_id);  
    ELSE  
        DBMS_OUTPUT.PUT_LINE('Bonus applied to ' || SQL%ROWCOUNT ||  
' employee(s) in department ' || p_department_id);  
    END IF;  
    COMMIT;  
END UpdateEmployeeBonus;
```

 Insert Into Editor

Result

Procedure UPDATEEMPLOYEEBONUS compiled

Scenario 3: Customers should be able to transfer funds between their accounts.

```

CREATE OR REPLACE PROCEDURE TransferFunds (
    p_from_account IN NUMBER,
    p_to_account   IN NUMBER,
    p_amount       IN NUMBER
)
AS
    v_balance NUMBER;
BEGIN
    -- Get current balance of source account
    SELECT balance INTO v_balance
    FROM accounts
    WHERE account_id = p_from_account;

    IF v_balance < p_amount THEN
        DBMS_OUTPUT.PUT_LINE('Transfer failed: insufficient balance');
    ELSE
        UPDATE accounts
        SET balance = balance - p_amount
        WHERE account_id = p_from_account;
        UPDATE accounts
        SET balance = balance + p_amount
        WHERE account_id = p_to_account;
        COMMIT;
        DBMS_OUTPUT.PUT_LINE('Transfer successful');
    END IF;
END TransferFunds;

```

Exercise 1: Setting Up JUnit

```
* Terminal will be reused by tasks, press any key to close it.

* Executing task: ""c:\Users\phius\.vscode\extensions\vscjava.vscode-maven-0.45.3\resources\maven-wrapper\mvnw.cmd" org.apache.mave
.plugins:maven-archetype-plugin:3.1.2:generate -DarchetypeArtifactId="maven-archetype-quickstart" -DarchetypeGroupId="org.apache.mav
n.archetypes" -DarchetypeVersion="1.4" -DgroupId="com.example" -DartifactId="demo" -DoutputDirectory="p:\Maven""
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.1.2:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.1.2:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.1.2:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Interactive mode
[INFO] Archetype repository not defined. Using the one from [org.apache.maven.archetypes:maven-archetype-quickstart:1.5] found in cat
log remote
[INFO] Using property: groupId = com.example
[INFO] Using property: artifactId = demo
[INFO] Define value for property 'version' 1.0-SNAPSHOT: :
[INFO] Using property: package = com.example
[INFO] Confirm properties configuration:
[INFO] groupId: com.example

BOX Agent Indexing completed. Java: Ready Generative Language Client Open Website Go Live BLACKBOXAI: Open Chat Prettier
```

EXPLORER

OPEN EDITORS

- ✖ pom.xml
- CalculatorTest.java src\test\java\com\example
- App.java src\main\java\com\example

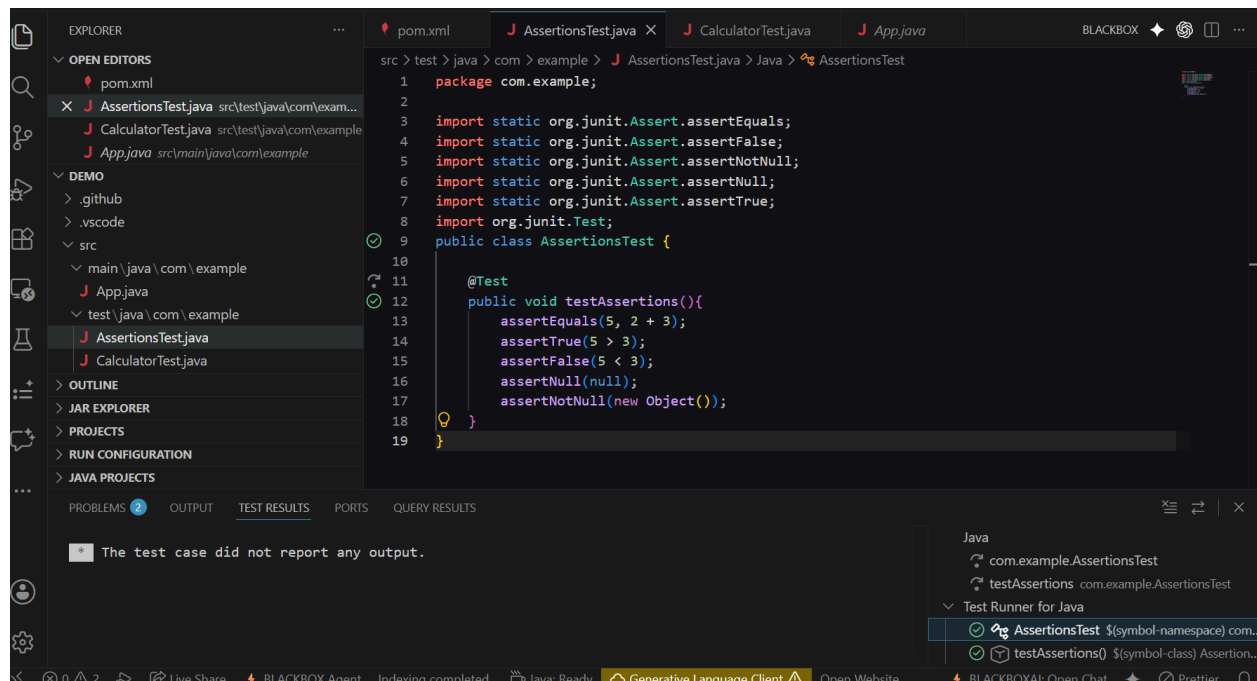
DEMO

- > .github
- > .vscode
- > src
 - main\java\com\example
 - App.java
 - test\java\com\example
 - CalculatorTest.java
 - > target
- ✖ pom.xml

pom.xml

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0"
3         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
5     <modelVersion>4.0.0</modelVersion>
6     <groupId>com.example</groupId>
7     <artifactId>demo</artifactId>
8     <version>1.0.0-SNAPSHOT</version>
9
10    <dependencies>
11        <dependency>
12            <groupId>junit</groupId>
13            <artifactId>junit</artifactId>
14            <version>4.13.2</version>
15            <scope>test</scope>
16        </dependency>
17    </dependencies>
18 </project>
```

Exercise 3: Assertions in JUnit



Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit